

## 1. Create a Java application to manage employee records using CRUD operations

Code:

```
import java.sql.*;
```

```
public class EmployeeCRUD {
```

```
    public static void main(String[] args) {
```

```
        String url = "jdbc:mysql://localhost:3306/SIT_CSE";
```

```
        String user = "root";
```

```
        String password = "root";
```

```
        try {
```

```
            // Load JDBC Driver
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            // Establish Connection
```

```
            Connection con = DriverManager.getConnection(url, user, password);
```

```
            System.out.println("Database Connected Successfully!");
```

```
            // CREATE - INSERT
```

```
            String insertQuery =
```

```
                "INSERT INTO employee VALUES (?, ?, ?, ?)";
```

```
            PreparedStatement psInsert =
```

```
                con.prepareStatement(insertQuery);
```

```
            psInsert.setInt(1, 104);
```

```
            psInsert.setString(2, "Akshat");
```

```
            psInsert.setString(3, "IT");
```

```
            psInsert.setDouble(4, 60000);
```

```
            psInsert.executeUpdate();
```

```
            System.out.println("Employee Inserted Successfully!");
```

```
            // READ - SELECT
```

```
            String selectQuery = "SELECT * FROM employee";
```

```
            Statement st = con.createStatement();
```

```
            ResultSet rs = st.executeQuery(selectQuery);
```

```
System.out.println("\n===== EMPLOYEE RECORDS =====");
```

```
while (rs.next()) {  
    System.out.println(  
        rs.getInt("Eid") + " | " +  
        rs.getString("Ename") + " | " +  
        rs.getString("Department") + " | " +  
        rs.getDouble("Salary")  
    );  
}
```

```
// UPDATE
```

```
String updateQuery =  
    "UPDATE employee SET Salary = ? WHERE Eid = ?";
```

```
PreparedStatement psUpdate =  
    con.prepareStatement(updateQuery);
```

```
psUpdate.setDouble(1, 65000);  
psUpdate.setInt(2, 104);
```

```
psUpdate.executeUpdate();
```

```
System.out.println("\nEmployee Updated Successfully!");
```

```
// DELETE
```

```
String deleteQuery =  
    "DELETE FROM employee WHERE Eid = ?";
```

```
PreparedStatement psDelete =  
    con.prepareStatement(deleteQuery);
```

```
psDelete.setInt(1, 103);
```

```
psDelete.executeUpdate();
```

```
System.out.println("Employee Deleted Successfully!");
```

```
// Close resources
```

```
rs.close();  
st.close();  
psInsert.close();  
psUpdate.close();  
psDelete.close();
```

```

        con.close();

    } catch (Exception e) {
        System.out.println(e);
    }
}
}

CREATE DATABASE SIT_CSE;
USE SIT_CSE;

CREATE TABLE employee (
    Eid INT PRIMARY KEY,
    Ename VARCHAR(100),
    Department VARCHAR(50),
    Salary DOUBLE
);

INSERT INTO employee VALUES
(101, 'Rahul', 'IT', 50000),
(102, 'Aman', 'HR', 45000),
(103, 'Rohit', 'Finance', 55000);

```

Output:

```

Database Connected Successfully!
Employee Inserted Successfully!

===== EMPLOYEE RECORDS =====
101 | Rahul | IT | 50000.0
102 | Aman | HR | 45000.0
103 | Rohit | Finance | 55000.0
104 | Akshat | IT | 60000.0

Employee Updated Successfully!
Employee Deleted Successfully!

```

2. Create a Java application to manage student records using CRUD operations. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database

```

import java.sql.*;

public class StudentCRUD {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/SIT_CSE";
        String user = "root";
        String password = "root";

        try {
            // Load JDBC Driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish Connection
            Connection con =
                DriverManager.getConnection(url, user, password);

            System.out.println("Database Connected Successfully!");

            // ===== CREATE =====

            String insertQuery =
                "INSERT INTO student VALUES (?, ?, ?, ?)";

            PreparedStatement psInsert =
                con.prepareStatement(insertQuery);

            psInsert.setInt(1, 4);
            psInsert.setString(2, "Akshat");
            psInsert.setString(3, "CSE");
            psInsert.setDouble(4, 90);

            psInsert.executeUpdate();

            System.out.println("Student Inserted Successfully!");

            // ===== READ =====

            String selectQuery =

```

```

        "SELECT * FROM student";

Statement st = con.createStatement();

ResultSet rs =
    st.executeQuery(selectQuery);

System.out.println("\n===== STUDENT RECORDS =====");

while (rs.next()) {

    System.out.println(
        "Roll No: " + rs.getInt("RollNo")
        + "\nName: " + rs.getString("Name")
        + "\nCourse: " + rs.getString("Course")
        + "\nMarks: " + rs.getDouble("Marks")
    );

    System.out.println("-----");
}

// ===== UPDATE =====

String updateQuery =
    "UPDATE student SET Marks = ? WHERE RollNo = ?";

PreparedStatement psUpdate =
    con.prepareStatement(updateQuery);

psUpdate.setDouble(1, 95);
psUpdate.setInt(2, 4);

psUpdate.executeUpdate();

System.out.println("Student Updated Successfully!");

// ===== DELETE =====

String deleteQuery =
    "DELETE FROM student WHERE RollNo = ?";

PreparedStatement psDelete =
    con.prepareStatement(deleteQuery);

```

```

        psDelete.setInt(1, 3);

        psDelete.executeUpdate();

        System.out.println("Student Deleted Successfully!");

        // Close resources
        rs.close();
        st.close();
        psInsert.close();
        psUpdate.close();
        psDelete.close();
        con.close();

    } catch (Exception e) {
        System.out.println(e);
    }
}
}

```

```
USE SIT_CSE;
```

```

CREATE TABLE student (
    RollNo INT PRIMARY KEY,
    Name VARCHAR(100),
    Course VARCHAR(50),
    Marks DOUBLE
);

```

```

INSERT INTO student VALUES
(1, 'Rahul', 'CSE', 78),
(2, 'Aman', 'CSE', 85),
(3, 'Rohit', 'IT', 72);

```

Output:

Database Connected Successfully!

Student Inserted Successfully!

===== STUDENT RECORDS =====

Roll No: 1

Name: Rahul

Course: CSE

Marks: 78.0

-----

Roll No: 2

Name: Aman

Course: CSE

Marks: 85.0

-----

Roll No: 3

Name: Rohit

Course: IT

Marks: 72.0

-----

Roll No: 4

Name: Akshat

Course: CSE

Marks: 90.0

-----

Student Updated Successfully!

Student Deleted Successfully!